# Block practice · Binary numbers and logic

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Binary to denary and back (unsigned, 0 to 255)

Complete a truth table

How many patterns from n bits

Adding binary numbers and overflow

Sign and magnitude numbers

Two's complement numbers

Write a logic statement for a problem

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# Practice workbook · Binary to denary and back (unsigned, 0 to 255)

Converting a whole number 0-255 to an 8-bit unsigned pattern, or an unsigned pattern (including a short sample value) to denary.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-01

# 1. Question 2 · part d.i · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part d.i only.

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**2** Computers manipulate binary patterns. People interpret those patterns.

(a) Complete the table by adding the hexadecimal notation for each of the denary values.

**(2)**

| Denary | Hexadecimal |
|---|---|
| 8 | |
| 12 | |

(b) Identify the expression to give the number of unique binary patterns that can be stored in six bits.

**(1)**

- ☒ **A** $6^2$
- ☒ **B** $6 \times 2$
- ☒ **C** $6^2 - 1$
- ☒ **D** $2^6$

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(c) Binary patterns are manipulated by shifts.

(i) Give the result of applying a logical shift right by two to the binary pattern 0101 1100

**(1)**

(ii) Give the result of applying an arithmetic shift right by three to the binary pattern 1100 0101

**(1)**

# 1. Question 2 · part d.i · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part d.i only.

(d) Binary patterns can be interpreted as signed or unsigned integers.

(i) Convert the denary unsigned integer 60 to 8-bit binary.

(2)

(ii) Here is a binary bit pattern for a signed integer in sign and magnitude format.

1001 0110

Convert the binary bit pattern to denary.

Be sure to include a sign symbol in your answer.

(2)

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6

P 7 2 9 3 7 A 0 6 2 4

## 1. Question 2 · part d.i · 2 marks · Working and answer

Working and answer

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# 2. Question 3 · part e.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part e.ii only.

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**3** Isra works in an office building where she has access to different types of network.

(a) Isra uses different types of networks for different tasks.

(i) Name the type of network she uses to print a document on the printer in the office next door.

(1)

(ii) Name the type of network she uses to order stationery from an online supplier.

(1)

(b) Identify the measurement used for network data speeds.

(1)

- ☒ **A** Megabits per second
- ☒ **B** Megapixels per second
- ☒ **C** Mebicycles per second
- ☒ **D** Mebihertz per second

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## 2. Question 3 · part e.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part e.ii only.

(c) Isra uses her tablet computer and smartphone to access email.

She wants to set up a new email account.

State the email protocol she should use.

Justify your choice.

(3)

Email protocol

Justification

(d) The table describes characteristics of different network topologies.

Complete the table to match each characteristic to **one** network topology.

(4)

| Characteristic | Bus | Ring | Star | Mesh |
|---|---|---|---|---|
| Network performance degrades as more devices are added. | ☒ | ☒ | ☒ | ☒ |
| All devices are directly connected to all others. | ☒ | ☒ | ☒ | ☒ |
| Each node has a physical attachment to a routing device. | ☒ | ☒ | ☒ | ☒ |
| There are no data collisions, because packets travel in the same direction. | ☒ | ☒ | ☒ | ☒ |

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## 2. Question 3 · part e.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part e.ii only.

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(e) Information sent across networks is represented in bit patterns.

(i) The bit pattern 1101 0001 uses sign and magnitude representation.

Convert this bit pattern to a denary number.

(2)

(ii) Convert the denary number 75 to 8-bit binary.

(2)

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## 2. Question 3 · part e.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part e.ii only.

(iii) The addition of these two 8-bit binary patterns generates an error condition.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | |
| 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | + |
| 0 | 0 | 1 | 0 | 1 | 1 | 0 | 1 | |

Explain this error condition.

(2)

(Total for Question 3 = 16 marks)

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Working and answer

# 3. Question 4 · part a.vi · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part a.vi only.

**4** Ibrahim works in the music industry.

(a) He is learning how sound is represented in binary.

**Figure 3** illustrates this process.

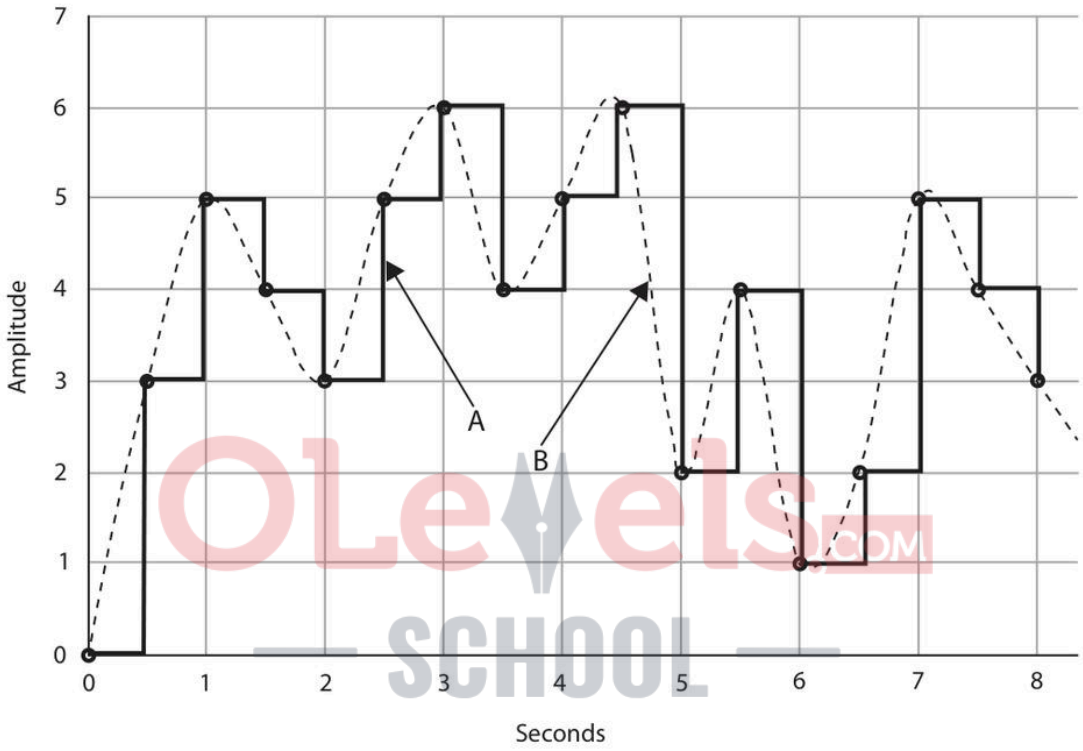



**Figure 3**

(i) State what the solid line labelled **A** represents.

**(1)**

(ii) State what the dashed line labelled **B** represents.

**(1)**

(iii) Give the sampling frequency, including the correct units.

**(1)**

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**10**

P 6 1 8 8 4 R R A 0 1 0 2 0

## 3. Question 4 · part a.vi · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part a.vi only.

(iv) Give **one** impact of increasing the sampling frequency.

**(1)**

(v) The bit-depth measures the resolution of the sound sample.

State the minimum bit-depth needed to store the sound sampled in **Figure 3**.

Justify your answer.

**(2)**

Bit-depth

Justification

(vi) Give the value in binary of the audio sample at the 4th second.

**(1)**

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## 3. Question 4 · part a.vi · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part a.vi only.

(b) Ibrahim wants to store a music file.

- The sampling frequency is 47 kHz.
- The bit-depth is 64
- The analogue signal lasts for 819 seconds.

Construct an expression to show how the file size, in bytes, is calculated.

You do **not** need to do the calculation.

**(4)**

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(c) Ibrahim wants to upload a music file to his online storage. The file has been compressed using a lossy algorithm.

Give **one** drawback of using a lossy algorithm for this purpose.

**(1)**

**(Total for Question 4 = 12 marks)**

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## 3. Question 4 · part a.vi · 1 mark · Working and answer

Working and answer

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## 4. Question 1 · part a.i · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part a.i only.

**1** Computers use binary to represent and store data.

(a) Binary and hexadecimal notation can be used to represent numbers.

(i) Convert the denary number 77 to 8-bit binary.

**(2)**

(ii) Convert the denary number –126 to 8-bit binary using two's complement.

**(2)**

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(iii) Convert the binary pattern 11000110 to hexadecimal.

**(2)**

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**2**

P 6 9 2 9 7 A 0 2 2 0

## 4. Question 1 · part a.i · 2 marks · Working and answer

Working and answer

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## 5. Question 2 · part a.i · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part a.i only.

**2** Computers use binary to represent and store data.

(a) The denary number 78 is the ASCII code for the character **N**.

(i) Convert the denary number 78 to 8-bit binary.

**(2)**

(ii) Identify the number of characters that can be represented using standard ASCII.

**(1)**

- ☒ **A** 64
- ☒ **B** 128
- ☒ **C** 256
- ☒ **D** 512

(iii) Explain **one** reason for using Unicode rather than ASCII to encode languages other than English.

**(2)**

(b) Convert the denary number –43 to 8-bit binary using sign and magnitude representation.

**(2)**

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## 5. Question 2 · part a.i · 2 marks · Working and answer

Working and answer

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## 6. Question 3 · part a.iii · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part a.iii only.

**3** Computers use binary to represent and store data.

(a) Two images are stored.

- Image 1 has a colour depth of 8 bits.
- Image 2 has a colour depth of 16 bits.

(i) State the number of colours that can be represented in Image 1.

**(1)**

(ii) Image 2 is 1920 pixels wide by 1080 pixels high.

Construct an expression to show the file size of the image in mebibytes.

You do **not** need to do the calculation.

**(3)**

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(iii) One pixel in Image 1 is represented by the unsigned binary integer

1100 1010

Identify the denary equivalent.

**(1)**

- ☒ **A** 54
- ☒ **B** 191
- ☒ **C** 202
- ☒ **D** –74

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## 6. Question 3 · part a.iii · 1 mark

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Further practice · Attempt part a.iii only.

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(b) A register holds the value 1100 0001 as an 8-bit binary value in two's complement.

(i) Convert the two's complement binary integer 1100 0001 to denary.

**(2)**

(ii) Give the result of applying an arithmetic shift left by one to the binary pattern 1100 0001

**(1)**

(iii) The value 1111 1111 is added to the value 1100 0001 stored in the register.

An error occurs.

Explain this error.

**(2)**

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## 6. Question 3 · part a.iii · 1 mark · Working and answer

Working and answer

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# Practice workbook · Complete a truth table

Filling in input combinations and intermediate and final columns for expressions using AND, OR and NOT.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-10

# 1. Question 4 · part c.i · 3 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part c.i only.

**4** A house has solar panels and a battery storage system.

(a) A smartphone application is used to monitor the battery storage system remotely. The smartphone uses a 5G connection.

(i) Give **two** benefits of a 5G connection.

**(2)**

1 .....

.....

2 .....

.....

(ii) The application requires users to authenticate their identity.

One method of authentication is a user ID and password.

Describe **one other** method the application could use.

**(2)**

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(iii) Software developers perform code reviews.

Give **two** reasons for carrying out a code review.

**(2)**

1 .....

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2 .....

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# 1. Question 4 · part c.i · 3 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part c.i only.

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(b) The battery storage system stores readings as signed 8-bit binary integers in **sign and magnitude** format.

Complete the table to show the 8-bit binary pattern and denary value for the lowest and the highest values that can be stored.

**(2)**

| | 8-bit binary pattern | Denary value |
|---|---|---|
| Lowest value | | |
| Highest value | | |

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# 1. Question 4 · part c.i · 3 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part c.i only.

(c) The solar panels and battery storage system are controlled by a program.

(i) An LED is displayed when the system is on (**O**), the sun is shining (**S**), and the house is not using all the power (**P**) generated by the solar panels.

Complete the truth table to show the results of each operation.

(3)

| O | S | P | O AND S | NOT P | (O AND S) AND (NOT P) |
|---|---|---|---|---|---|
| 0 | 0 | 0 | | | |
| 0 | 0 | 1 | | | |
| 0 | 1 | 0 | | | |
| 0 | 1 | 1 | | | |
| 1 | 0 | 0 | | | |
| 1 | 0 | 1 | | | |
| 1 | 1 | 0 | | | |
| 1 | 1 | 1 | | | |

(ii) The start-up instructions for the solar panels and battery storage system are stored in read-only memory (ROM).

Explain why the start-up instructions are stored in ROM.

(2)

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**(Total for Question 4 = 13 marks)**

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## 1. Question 4 · part c.i · 3 marks · Working and answer

Working and answer

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## 2. Question 5 · part a · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

**5** Programmers work as part of a team to develop software.

(a) They use truth tables to determine if an expression evaluates to true or false.

Complete this truth table for the Boolean expression

**P AND (Q OR R)**

You may not need to use all rows.

**(3)**

| P | Q | R | Q OR R | P AND (Q OR R) |
|---|---|---|---|---|
| | | | | |
| | | | | |
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## 2. Question 5 · part a · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

(b) The team is working on software for a ride-on lawn mower.

The lawn mower must not start unless it is safe for the driver.

- The brake (B) must be engaged.
- The driver's seat sensor (S) must be engaged.
- The blade safety lock (L) must be removed.

Construct a Boolean expression, using AND, OR, and NOT with the letters B, S and L to show the safe starting condition.

**(2)**

(c) The team uses code reviews as part of the software development process.

State **two** ways in which code reviews can help protect software systems from cyber attacks.

**(2)**

1 .....

2 .....

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## 2. Question 5 · part a · 3 marks · Working and answer

Working and answer

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## 3. Question 4 · part c · 3 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

**4** Scientists often use models and simulations.

(a) Give **one** reason, other than cost, why scientists use simulations.

**(1)**

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.....

(b) Explain **one** reason why a simulation may require a powerful processor.

**(2)**

.....

.....

.....

.....

(c) A scientist uses Boolean logic in programs.

Complete the table to show the results of each operation.

**(3)**

| R | S | W | NOT S | R AND W | (NOT S) OR (R AND W) |
|---|---|---|---|---|---|
| 0 | 0 | 0 | | | |
| 0 | 0 | 1 | | | |
| 0 | 1 | 0 | | | |
| 0 | 1 | 1 | | | |
| 1 | 0 | 0 | | | |
| 1 | 0 | 1 | | | |
| 1 | 1 | 0 | | | |
| 1 | 1 | 1 | | | |

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## 3. Question 4 · part c · 3 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

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(d) Von Neumann developed the stored program concept.

(i) Describe the **stored program concept**.

**(2)**

(ii) The fetch-decode-execute cycle is the cycle the central processing unit (CPU) follows in order to process instructions.

Name **two** registers used in the cycle.

**(2)**

1 .....

2 .....

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(iii) Describe the role of the address bus and the data bus during the **fetch** part of the cycle.

**(2)**

(iv) Complete the table to show the effect on a computer system of increasing the width of a bus.

**(2)**

| Change made | Effect |
|---|---|
| Increasing the width of the data bus | |
| Increasing the width of the address bus | |

**(Total for Question 4 = 14 marks)**

## 3. Question 4 · part c · 3 marks · Working and answer

Working and answer

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# 4. Question 5 · part d.i · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part d.i only.

**5** Hardware devices execute programs to carry out a variety of tasks.

(a) A program controls a bee character in an animation.

The bee can turn to face North, East, South or West.

The bee can move any number of steps in the direction it is facing.

Complete the table to show one input and one output required to move the bee.

**(2)**

| Requirement | Example |
|---|---|
| Input | |
| Process | Calculate the path the bee will move along to its new position |
| Output | |

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# 4. Question 5 · part d.i · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part d.i only.

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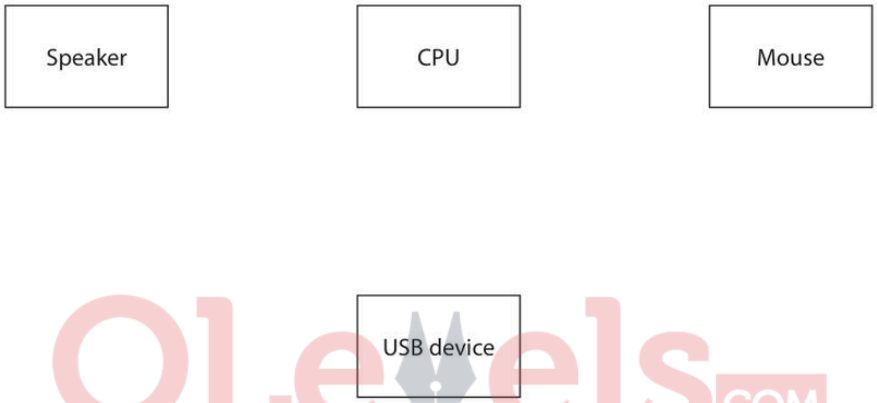
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(b) Computers are made up of hardware components.

(i) Complete the diagram by adding directional arrows between the components to show the flow of communication.

**(3)**




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# 4. Question 5 · part d.i · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part d.i only.

(ii) Identify what is stored in ROM.

(1)

- ☒ **A** The software firewall
- ☒ **B** The basic input output system
- ☒ **C** The operating system
- ☒ **D** The user interface code

(iii) Cache is used as temporary storage.

One type of cache is located between main memory and the CPU.

Explain **one** reason cache is used in a computer.

(2)

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# 4. Question 5 · part d.i · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part d.i only.

(c) A washing machine uses several different embedded systems.

One embedded system uses a switch to identify the type of wash cycle selected by the user.

Describe **one other** example of an embedded system found in a washing machine.

**(2)**

(d) Programs use logic statements to control physical hardware.

(i) A window shuts when the temperature is too cool or it is a rainy night.

The values are defined as:

- A shows it is night time
- B shows it is too cool
- C shows it is raining.

Complete the truth table to show the results of each operation.

Two rows have been done for you.

**(3)**

| A | B | C | A AND C | (A AND C) OR B |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | | |
| 0 | 1 | 0 | | |
| 0 | 1 | 1 | | |
| 1 | 0 | 0 | | |
| 1 | 0 | 1 | | |
| 1 | 1 | 0 | | |
| 1 | 1 | 1 | 1 | 1 |

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## 4. Question 5 · part d.i · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part d.i only.

(ii) A warehouse has an automated alarm system.

When the alarm system is activated it will sound if:

- a movement sensor (M) is activated
- a pressure pad (P) is activated
- a key code (C) to deactivate the alarm system has not been entered.

Construct a logic statement, using AND, OR and NOT with the letters M, P and C, to show the conditions that will sound the alarm.

**(3)**

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## 4. Question 5 · part d.i · 3 marks · Working and answer

Working and answer

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## 5. Question 3 · part a · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part a only.

**3** Programs use logic, display data and validate input.

(a) The truth table shows the initial values for A and B.

Complete the truth table to show the results of each operation.

**(4)**

| A | B | A OR B | NOT B | (A OR B) AND (NOT B) |
|---|---|---|---|---|
| 0 | 0 | | | |
| 0 | 1 | | | |
| 1 | 0 | | | |
| 1 | 1 | | | |

(b) **Figure 2** shows the intended output from a program that displays every other name in an array of star names.

```
Alasia
Castor
Electra
Gudja
Izar
Kang
Maia
Ogma
```

**Figure 2**

Open **Q03b** in the code editor.

Amend the code to display every other name in the array of star names.

Do not add any further functionality.

Save your code as **Q03bFINISHED** with the correct file extension for the programming language.

**(3)**

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P 7 5 7 3 7 A 0 6 1 6

## 5. Question 3 · part a · 4 marks · Working and answer

Working and answer

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# Practice workbook · How many patterns from n bits

Linking a number of bits to the number of distinct patterns/colours/levels, in both directions, including assigning unique codes with the minimum colour depth.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-08

# 1. Question 2 · part d.i · 2 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part d.i only.

**2** Computers use binary to represent and store data.

(a) The denary number 78 is the ASCII code for the character **N**.

(i) Convert the denary number 78 to 8-bit binary.

**(2)**

(ii) Identify the number of characters that can be represented using standard ASCII.

**(1)**

- ☒ **A** 64
- ☒ **B** 128
- ☒ **C** 256
- ☒ **D** 512

(iii) Explain **one** reason for using Unicode rather than ASCII to encode languages other than English.

**(2)**

(b) Convert the denary number –43 to 8-bit binary using sign and magnitude representation.

**(2)**

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# 1. Question 2 · part d.i · 2 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part d.i only.

(c) Complete the table by adding these two 8-bit binary integers.

**(2)**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 | 1 | 0 | 0 |
| 0 | 0 | 0 | 1 | 0 | 1 | 1 | 0 |
| | | | | | | | |

(d) A bitmap image is made up of pixels.

(i) An image has five colours.

Complete the table by adding a unique binary pattern for each colour.

Each pattern must use the **same minimum colour depth**.

**(2)**

| Colour | Binary pattern |
|---|---|
| Green | |
| Black | |
| White | |
| Red | |
| Blue | |

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P 7 2 5 3 8 A 0 5 2 0

5

*Turn over* ▶

# 1. Question 2 · part d.i · 2 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part d.i only.

(ii) Another image is 3579 pixels high and 6128 pixels wide.

The image is stored with a 32-bit colour depth.

The metadata for the image is 732 bytes.

Construct an expression to show how the file size, in **megabytes**, is calculated.

You do **not** need to do the calculation.

**(4)**

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**(Total for Question 2 = 15 marks)**

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Working and answer

## 2. Question 4 · part a.v · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part a.v only.

**4** Ibrahim works in the music industry.

(a) He is learning how sound is represented in binary.

**Figure 3** illustrates this process.

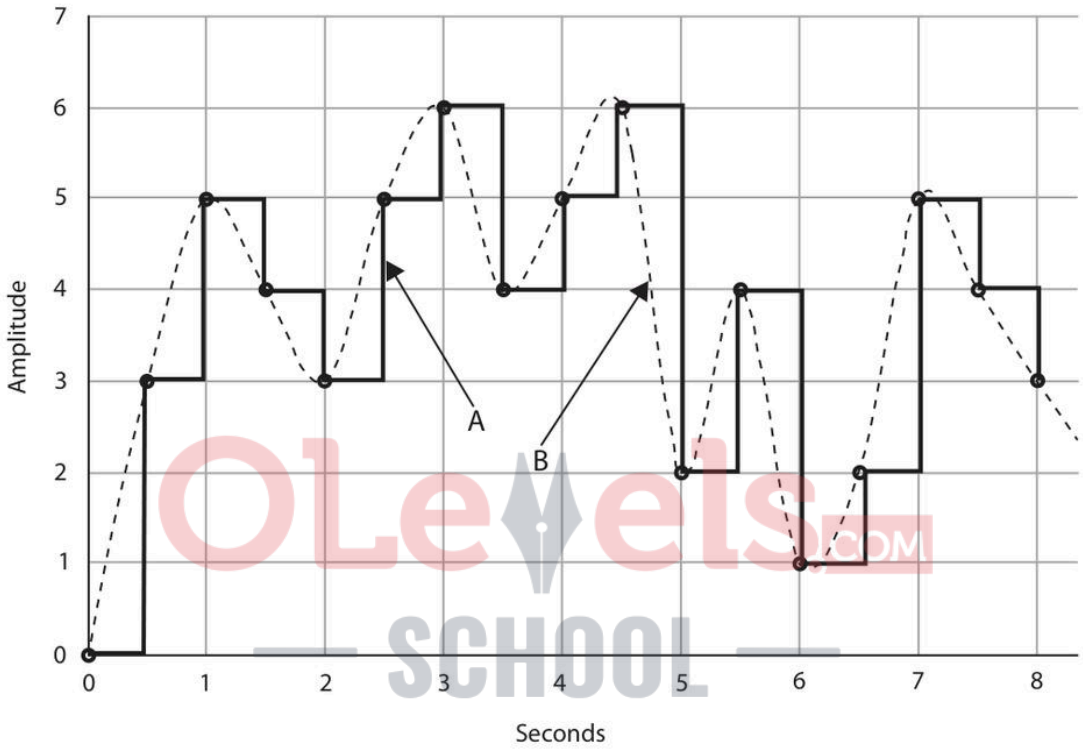



**Figure 3**

(i) State what the solid line labelled **A** represents.

**(1)**

(ii) State what the dashed line labelled **B** represents.

**(1)**

(iii) Give the sampling frequency, including the correct units.

**(1)**

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P 6 1 8 8 4 R R A 0 1 0 2 0

## 2. Question 4 · part a.v · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part a.v only.

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(iv) Give **one** impact of increasing the sampling frequency.

(1)

(v) The bit-depth measures the resolution of the sound sample.

State the minimum bit-depth needed to store the sound sampled in **Figure 3**.

Justify your answer.

(2)

Bit-depth

Justification

(vi) Give the value in binary of the audio sample at the 4th second.

(1)

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## 2. Question 4 · part a.v · 2 marks · Working and answer

Working and answer

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## 3. Question 1 · part c · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part c only.

**1** Binary digits (bits) are grouped together to represent different types of data.

(a) Hexadecimal notation is sometimes used to represent patterns of binary digits.

Identify **one** reason why programmers use hexadecimal notation.

(1)

- ☒ **A** Easier for a computer to understand.
- ☒ **B** Easier for humans to read.
- ☒ **C** Takes up less computer memory.
- ☒ **D** Quicker than binary to execute.

(b) Convert the bit pattern 0101 1010 to hexadecimal.

(2)

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(c) Identify the number of binary patterns that can be represented by 8 bits.

(1)

- ☒ **A** 128
- ☒ **B** 256
- ☒ **C** 512
- ☒ **D** 1024

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## 3. Question 1 · part c · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part c only.

(d) Complete the table by adding these two 8-bit binary integers.

**(2)**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| | | | | | | | |

(e) Give the result of performing a logical left shift of 1 place on the binary integer 0100.

**(1)**

(f) Give the number of bits per character used by standard ASCII.

**(1)**

(g) **Figure 1** shows a black and white bitmap image.

The pixels in row 5 are represented by the binary pattern 1001 1001

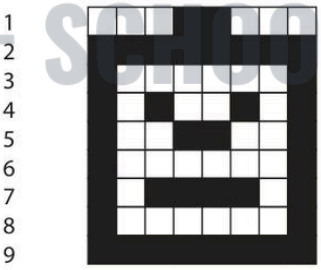



**Figure 1**

(i) State what is meant by the term **pixel**.

**(1)**

(ii) Construct an expression to show the number of pixels in the image.

**(1)**

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## 3. Question 1 · part c · 1 mark · Working and answer

Working and answer

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# 4. Question 1 · part g.iii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part g.iii only.

**1** Binary digits (bits) are grouped together to represent different types of data.

(a) Hexadecimal notation is sometimes used to represent patterns of binary digits.

Identify **one** reason why programmers use hexadecimal notation.

(1)

- ☒ **A** Easier for a computer to understand.
- ☒ **B** Easier for humans to read.
- ☒ **C** Takes up less computer memory.
- ☒ **D** Quicker than binary to execute.

(b) Convert the bit pattern 0101 1010 to hexadecimal.

(2)

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(c) Identify the number of binary patterns that can be represented by 8 bits.

(1)

- ☒ **A** 128
- ☒ **B** 256
- ☒ **C** 512
- ☒ **D** 1024

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# 4. Question 1 · part g.iii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part g.iii only.

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(d) Complete the table by adding these two 8-bit binary integers.

**(2)**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| | | | | | | | |

(e) Give the result of performing a logical left shift of 1 place on the binary integer 0100.

**(1)**

(f) Give the number of bits per character used by standard ASCII.

**(1)**

(g) **Figure 1** shows a black and white bitmap image.

The pixels in row 5 are represented by the binary pattern 1001 1001

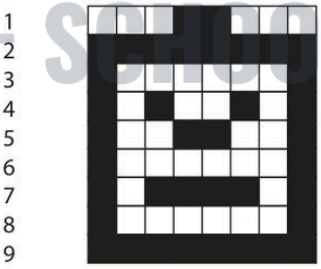



**Figure 1**

(i) State what is meant by the term **pixel**.

**(1)**

(ii) Construct an expression to show the number of pixels in the image.

**(1)**

## 4. Question 1 · part g.iii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part g.iii only.

(iii) The image is changed so that any pixel can be one of 16 different colours.

State the minimum number of bits that would be needed to represent **one** pixel.

(1)

.....

**(Total for Question 1 = 11 marks)**

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Working and answer

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## 5. Question 2 · part b · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part b only.

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**2** Computers manipulate binary patterns. People interpret those patterns.

(a) Complete the table by adding the hexadecimal notation for each of the denary values.

**(2)**

| Denary | Hexadecimal |
|---|---|
| 8 | |
| 12 | |

(b) Identify the expression to give the number of unique binary patterns that can be stored in six bits.

**(1)**

- ☒ **A** $6^2$
- ☒ **B** $6 \times 2$
- ☒ **C** $6^2 - 1$
- ☒ **D** $2^6$

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(c) Binary patterns are manipulated by shifts.

(i) Give the result of applying a logical shift right by two to the binary pattern 0101 1100

**(1)**

(ii) Give the result of applying an arithmetic shift right by three to the binary pattern 1100 0101

**(1)**

## 5. Question 2 · part b · 1 mark · Working and answer

Working and answer

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## 6. Question 3 · part a.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part a.i only.

**3** Computers use binary to represent and store data.

(a) Two images are stored.

- Image 1 has a colour depth of 8 bits.
- Image 2 has a colour depth of 16 bits.

(i) State the number of colours that can be represented in Image 1.

**(1)**

.....

(ii) Image 2 is 1920 pixels wide by 1080 pixels high.

Construct an expression to show the file size of the image in mebibytes.

You do **not** need to do the calculation.

**(3)**

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(iii) One pixel in Image 1 is represented by the unsigned binary integer

1100 1010

Identify the denary equivalent.

**(1)**

☒ **A** 54

☒ **B** 191

☒ **C** 202

☒ **D** –74

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## 6. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

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# Practice workbook · Adding binary numbers and overflow

Column addition of two 8-bit binary integers and explaining the overflow error when the result does not fit.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-06

# 1. Question 2 · part c · 2 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part c only.

**2** Computers use binary to represent and store data.

(a) The denary number 78 is the ASCII code for the character **N**.

(i) Convert the denary number 78 to 8-bit binary.

**(2)**

(ii) Identify the number of characters that can be represented using standard ASCII.

**(1)**

- ☒ **A** 64
- ☒ **B** 128
- ☒ **C** 256
- ☒ **D** 512

(iii) Explain **one** reason for using Unicode rather than ASCII to encode languages other than English.

**(2)**

(b) Convert the denary number –43 to 8-bit binary using sign and magnitude representation.

**(2)**

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# 1. Question 2 · part c · 2 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part c only.

(c) Complete the table by adding these two 8-bit binary integers.

**(2)**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 | 1 | 0 | 0 |
| 0 | 0 | 0 | 1 | 0 | 1 | 1 | 0 |
| | | | | | | | |

(d) A bitmap image is made up of pixels.

(i) An image has five colours.

Complete the table by adding a unique binary pattern for each colour.

Each pattern must use the **same minimum colour depth**.

**(2)**

| Colour | Binary pattern |
|---|---|
| Green | |
| Black | |
| White | |
| Red | |
| Blue | |

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## 1. Question 2 · part c · 2 marks · Working and answer

Working and answer

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## 2. Question 3 · part e.iii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part e.iii only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

**3** Isra works in an office building where she has access to different types of network.

(a) Isra uses different types of networks for different tasks.

(i) Name the type of network she uses to print a document on the printer in the office next door.

(1)

.....

.....

(ii) Name the type of network she uses to order stationery from an online supplier.

(1)

.....

.....

(b) Identify the measurement used for network data speeds.

(1)

- ☒ **A** Megabits per second
- ☒ **B** Megapixels per second
- ☒ **C** Mebicycles per second
- ☒ **D** Mebihertz per second

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## 2. Question 3 · part e.iii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part e.iii only.

(c) Isra uses her tablet computer and smartphone to access email.

She wants to set up a new email account.

State the email protocol she should use.

Justify your choice.

(3)

Email protocol

Justification

(d) The table describes characteristics of different network topologies.

Complete the table to match each characteristic to **one** network topology.

(4)

| Characteristic | Bus | Ring | Star | Mesh |
|---|---|---|---|---|
| Network performance degrades as more devices are added. | ☒ | ☒ | ☒ | ☒ |
| All devices are directly connected to all others. | ☒ | ☒ | ☒ | ☒ |
| Each node has a physical attachment to a routing device. | ☒ | ☒ | ☒ | ☒ |
| There are no data collisions, because packets travel in the same direction. | ☒ | ☒ | ☒ | ☒ |

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8

P 6 1 8 8 0 A 0 8 2 4

## 2. Question 3 · part e.iii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part e.iii only.

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(e) Information sent across networks is represented in bit patterns.

(i) The bit pattern 1101 0001 uses sign and magnitude representation.

Convert this bit pattern to a denary number.

(2)

(ii) Convert the denary number 75 to 8-bit binary.

(2)

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## 2. Question 3 · part e.iii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part e.iii only.

(iii) The addition of these two 8-bit binary patterns generates an error condition.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | |
| 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | + |
| 0 | 0 | 1 | 0 | 1 | 1 | 0 | 1 | |

Explain this error condition.

(2)

(Total for Question 3 = 16 marks)

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Working and answer

## 3. Question 1 · part d · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part d only.

**1** Binary digits (bits) are grouped together to represent different types of data.

(a) Hexadecimal notation is sometimes used to represent patterns of binary digits.

Identify **one** reason why programmers use hexadecimal notation.

(1)

- ☒ **A** Easier for a computer to understand.
- ☒ **B** Easier for humans to read.
- ☒ **C** Takes up less computer memory.
- ☒ **D** Quicker than binary to execute.

(b) Convert the bit pattern 0101 1010 to hexadecimal.

(2)

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(c) Identify the number of binary patterns that can be represented by 8 bits.

(1)

- ☒ **A** 128
- ☒ **B** 256
- ☒ **C** 512
- ☒ **D** 1024

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## 3. Question 1 · part d · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part d only.

(d) Complete the table by adding these two 8-bit binary integers.

**(2)**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| | | | | | | | |

(e) Give the result of performing a logical left shift of 1 place on the binary integer 0100.

**(1)**

(f) Give the number of bits per character used by standard ASCII.

**(1)**

(g) **Figure 1** shows a black and white bitmap image.

The pixels in row 5 are represented by the binary pattern 1001 1001

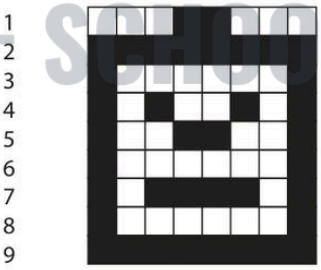



**Figure 1**

(i) State what is meant by the term **pixel**.

**(1)**

(ii) Construct an expression to show the number of pixels in the image.

**(1)**

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## 3. Question 1 · part d · 2 marks · Working and answer

Working and answer

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# 4. Question 3 · part b.iii · 2 marks

2024 JUN · Paper 01

Further practice · Attempt part b.iii only.

**3** Computers use binary to represent and store data.

(a) Two images are stored.

- Image 1 has a colour depth of 8 bits.
- Image 2 has a colour depth of 16 bits.

(i) State the number of colours that can be represented in Image 1.

**(1)**

.....

(ii) Image 2 is 1920 pixels wide by 1080 pixels high.

Construct an expression to show the file size of the image in mebibytes.

You do **not** need to do the calculation.

**(3)**

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(iii) One pixel in Image 1 is represented by the unsigned binary integer

1100 1010

Identify the denary equivalent.

**(1)**

☒ **A** 54

☒ **B** 191

☒ **C** 202

☒ **D** –74

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# 4. Question 3 · part b.iii · 2 marks

2024 JUN · Paper 01

Further practice · Attempt part b.iii only.

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(b) A register holds the value 1100 0001 as an 8-bit binary value in two's complement.

(i) Convert the two's complement binary integer 1100 0001 to denary.

(2)

(ii) Give the result of applying an arithmetic shift left by one to the binary pattern 1100 0001

(1)

(iii) The value 1111 1111 is added to the value 1100 0001 stored in the register.

An error occurs.

Explain this error.

(2)

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# 4. Question 3 · part b.iii · 2 marks

2024 JUN · Paper 01

Further practice · Attempt part b.iii only.

(c) The ASCII code for the character **1** is 011 0001

(i) Give the ASCII code for the character **5**

**(1)**

.....

(ii) Give **two** benefits of using ASCII to encode characters.

**(2)**

1 .....

.....

2 .....

.....

**(Total for Question 3 = 13 marks)**

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Working and answer

# Practice workbook · Sign and magnitude numbers

Signed integers held in 8-bit sign and magnitude: pattern to denary, denary to pattern, and the lowest/highest storable values.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-02

# 1. Question 2 · part b · 2 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part b only.

**2** Computers use binary to represent and store data.

(a) The denary number 78 is the ASCII code for the character **N**.

(i) Convert the denary number 78 to 8-bit binary.

**(2)**

(ii) Identify the number of characters that can be represented using standard ASCII.

**(1)**

- ☒ **A** 64
- ☒ **B** 128
- ☒ **C** 256
- ☒ **D** 512

(iii) Explain **one** reason for using Unicode rather than ASCII to encode languages other than English.

**(2)**

(b) Convert the denary number –43 to 8-bit binary using sign and magnitude representation.

**(2)**

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P 7 2 5 3 8 A 0 4 2 0

# 1. Question 2 · part b · 2 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part b only.

(c) Complete the table by adding these two 8-bit binary integers.

**(2)**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 | 1 | 0 | 0 |
| 0 | 0 | 0 | 1 | 0 | 1 | 1 | 0 |
| | | | | | | | |

(d) A bitmap image is made up of pixels.

(i) An image has five colours.

Complete the table by adding a unique binary pattern for each colour.

Each pattern must use the **same minimum colour depth**.

**(2)**

| Colour | Binary pattern |
|---|---|
| Green | |
| Black | |
| White | |
| Red | |
| Blue | |

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## 1. Question 2 · part b · 2 marks · Working and answer

Working and answer

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# 2. Question 3 · part e.i · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part e.i only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

**3** Isra works in an office building where she has access to different types of network.

(a) Isra uses different types of networks for different tasks.

(i) Name the type of network she uses to print a document on the printer in the office next door.

(1)

(ii) Name the type of network she uses to order stationery from an online supplier.

(1)

(b) Identify the measurement used for network data speeds.

(1)

- ☒ **A** Megabits per second
- ☒ **B** Megapixels per second
- ☒ **C** Mebicycles per second
- ☒ **D** Mebihertz per second

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# 2. Question 3 · part e.i · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part e.i only.

(c) Isra uses her tablet computer and smartphone to access email.

She wants to set up a new email account.

State the email protocol she should use.

Justify your choice.

(3)

Email protocol

Justification

(d) The table describes characteristics of different network topologies.

Complete the table to match each characteristic to **one** network topology.

(4)

| Characteristic | Bus | Ring | Star | Mesh |
|---|---|---|---|---|
| Network performance degrades as more devices are added. | ☒ | ☒ | ☒ | ☒ |
| All devices are directly connected to all others. | ☒ | ☒ | ☒ | ☒ |
| Each node has a physical attachment to a routing device. | ☒ | ☒ | ☒ | ☒ |
| There are no data collisions, because packets travel in the same direction. | ☒ | ☒ | ☒ | ☒ |

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## 2. Question 3 · part e.i · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part e.i only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

(e) Information sent across networks is represented in bit patterns.

(i) The bit pattern 1101 0001 uses sign and magnitude representation.

Convert this bit pattern to a denary number.

(2)

(ii) Convert the denary number 75 to 8-bit binary.

(2)

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## 2. Question 3 · part e.i · 2 marks · Working and answer

Working and answer

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## 3. Question 2 · part a.i · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part a.i only.

**2** Computers use binary digits to represent data.

(a) Sign and magnitude and two's complement are two ways of representing signed integer numbers.

(i) Convert the denary number -94 to a binary pattern using sign and magnitude representation.

**(2)**

(ii) 1000 1001 is a signed integer that uses two's complement representation.

Convert this signed integer to denary.

**(2)**

(b) Denary numbers use base 10

Identify the base for hexadecimal numbers.

**(1)**

- ☒ **A** 32
- ☒ **B** 16
- ☒ **C** 8
- ☒ **D** 2

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## 3. Question 2 · part a.i · 2 marks · Working and answer

Working and answer

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# 4. Question 2 · part d.ii · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part d.ii only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

**2** Computers manipulate binary patterns. People interpret those patterns.

(a) Complete the table by adding the hexadecimal notation for each of the denary values.

**(2)**

| Denary | Hexadecimal |
|---|---|
| 8 | |
| 12 | |

(b) Identify the expression to give the number of unique binary patterns that can be stored in six bits.

**(1)**

- ☒ **A** $6^2$
- ☒ **B** $6 \times 2$
- ☒ **C** $6^2 - 1$
- ☒ **D** $2^6$

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(c) Binary patterns are manipulated by shifts.

(i) Give the result of applying a logical shift right by two to the binary pattern 0101 1100

**(1)**

(ii) Give the result of applying an arithmetic shift right by three to the binary pattern 1100 0101

**(1)**

# 4. Question 2 · part d.ii · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part d.ii only.

(d) Binary patterns can be interpreted as signed or unsigned integers.

(i) Convert the denary unsigned integer 60 to 8-bit binary.

(2)

(ii) Here is a binary bit pattern for a signed integer in sign and magnitude format.

1001 0110

Convert the binary bit pattern to denary.

Be sure to include a sign symbol in your answer.

(2)

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DO NOT WRITE IN THIS AREA

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6

P 7 2 9 3 7 A 0 6 2 4

# 4. Question 2 · part d.ii · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part d.ii only.

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(iii) Negating a signed integer means changing its sign without changing its value.

The negation of +16 is –16. The negation of –24 is +24.

Here is the binary bit pattern for a signed integer in two's complement format.

1110 0101

Convert the binary pattern to its negation in two's complement.

**(2)**

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(e) Construct an expression to convert 13 kilobytes to kibibytes.

You **do not** need to do the calculation.

**(2)**

**(Total for Question 2 = 13 marks)**

## 4. Question 2 · part d.ii · 2 marks · Working and answer

Working and answer

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## 5. Question 4 · part b · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b only.

**4** A house has solar panels and a battery storage system.

(a) A smartphone application is used to monitor the battery storage system remotely. The smartphone uses a 5G connection.

(i) Give **two** benefits of a 5G connection.

**(2)**

1 .....

.....

2 .....

.....

(ii) The application requires users to authenticate their identity.

One method of authentication is a user ID and password.

Describe **one other** method the application could use.

**(2)**

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(iii) Software developers perform code reviews.

Give **two** reasons for carrying out a code review.

**(2)**

1 .....

.....

2 .....

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## 5. Question 4 · part b · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b only.

(b) The battery storage system stores readings as signed 8-bit binary integers in **sign and magnitude** format.

Complete the table to show the 8-bit binary pattern and denary value for the lowest and the highest values that can be stored.

**(2)**

| | 8-bit binary pattern | Denary value |
|---|---|---|
| Lowest value | | |
| Highest value | | |

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## 5. Question 4 · part b · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b only.

(c) The solar panels and battery storage system are controlled by a program.

(i) An LED is displayed when the system is on (**O**), the sun is shining (**S**), and the house is not using all the power (**P**) generated by the solar panels.

Complete the truth table to show the results of each operation.

(3)

| O | S | P | O AND S | NOT P | (O AND S) AND (NOT P) |
|---|---|---|---|---|---|
| 0 | 0 | 0 | | | |
| 0 | 0 | 1 | | | |
| 0 | 1 | 0 | | | |
| 0 | 1 | 1 | | | |
| 1 | 0 | 0 | | | |
| 1 | 0 | 1 | | | |
| 1 | 1 | 0 | | | |
| 1 | 1 | 1 | | | |

(ii) The start-up instructions for the solar panels and battery storage system are stored in read-only memory (ROM).

Explain why the start-up instructions are stored in ROM.

(2)

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**(Total for Question 4 = 13 marks)**

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## 5. Question 4 · part b · 2 marks · Working and answer

Working and answer

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# Practice workbook · Two's complement numbers

8-bit two's complement integers: pattern to denary, negative denary to pattern, and negating a pattern.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-03

# 1. Question 3 · part b.i · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part b.i only.

**3** Computers use binary to represent and store data.

(a) Two images are stored.

- Image 1 has a colour depth of 8 bits.
- Image 2 has a colour depth of 16 bits.

(i) State the number of colours that can be represented in Image 1.

**(1)**

.....

(ii) Image 2 is 1920 pixels wide by 1080 pixels high.

Construct an expression to show the file size of the image in mebibytes.

You do **not** need to do the calculation.

**(3)**

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(iii) One pixel in Image 1 is represented by the unsigned binary integer

1100 1010

Identify the denary equivalent.

**(1)**

- ☒ **A** 54
- ☒ **B** 191
- ☒ **C** 202
- ☒ **D** –74

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# 1. Question 3 · part b.i · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part b.i only.

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(b) A register holds the value 1100 0001 as an 8-bit binary value in two's complement.

(i) Convert the two's complement binary integer 1100 0001 to denary.

**(2)**

(ii) Give the result of applying an arithmetic shift left by one to the binary pattern 1100 0001

**(1)**

(iii) The value 1111 1111 is added to the value 1100 0001 stored in the register.

An error occurs.

Explain this error.

**(2)**

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## 1. Question 3 · part b.i · 2 marks · Working and answer

Working and answer

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## 2. Question 2 · part a.ii · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part a.ii only.

**2** Computers use binary digits to represent data.

(a) Sign and magnitude and two's complement are two ways of representing signed integer numbers.

(i) Convert the denary number -94 to a binary pattern using sign and magnitude representation.

**(2)**

(ii) 1000 1001 is a signed integer that uses two's complement representation.

Convert this signed integer to denary.

**(2)**

(b) Denary numbers use base 10

Identify the base for hexadecimal numbers.

**(1)**

- ☒ **A** 32
- ☒ **B** 16
- ☒ **C** 8
- ☒ **D** 2

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## 2. Question 2 · part a.ii · 2 marks · Working and answer

Working and answer

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## 3. Question 1 · part a.ii · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part a.ii only.

**1** Computers use binary to represent and store data.

(a) Binary and hexadecimal notation can be used to represent numbers.

(i) Convert the denary number 77 to 8-bit binary.

**(2)**

(ii) Convert the denary number –126 to 8-bit binary using two's complement.

**(2)**

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(iii) Convert the binary pattern 11000110 to hexadecimal.

**(2)**

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## 3. Question 1 · part a.ii · 2 marks · Working and answer

Working and answer

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# 4. Question 2 · part d.iii · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part d.iii only.

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**2** Computers manipulate binary patterns. People interpret those patterns.

(a) Complete the table by adding the hexadecimal notation for each of the denary values.

**(2)**

| Denary | Hexadecimal |
|---|---|
| 8 | |
| 12 | |

(b) Identify the expression to give the number of unique binary patterns that can be stored in six bits.

**(1)**

- ☒ **A** $6^2$
- ☒ **B** $6 \times 2$
- ☒ **C** $6^2 - 1$
- ☒ **D** $2^6$

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(c) Binary patterns are manipulated by shifts.

(i) Give the result of applying a logical shift right by two to the binary pattern 0101 1100

**(1)**

(ii) Give the result of applying an arithmetic shift right by three to the binary pattern 1100 0101

**(1)**

# 4. Question 2 · part d.iii · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part d.iii only.

(d) Binary patterns can be interpreted as signed or unsigned integers.

(i) Convert the denary unsigned integer 60 to 8-bit binary.

(2)

(ii) Here is a binary bit pattern for a signed integer in sign and magnitude format.

1001 0110

Convert the binary bit pattern to denary.

Be sure to include a sign symbol in your answer.

(2)

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6

P 7 2 9 3 7 A 0 6 2 4

# 4. Question 2 · part d.iii · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part d.iii only.

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(iii) Negating a signed integer means changing its sign without changing its value.

The negation of +16 is –16. The negation of –24 is +24.

Here is the binary bit pattern for a signed integer in two's complement format.

1110 0101

Convert the binary pattern to its negation in two's complement.

**(2)**

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(e) Construct an expression to convert 13 kilobytes to kibibytes.

You **do not** need to do the calculation.

**(2)**

**(Total for Question 2 = 13 marks)**

## 4. Question 2 · part d.iii · 2 marks · Working and answer

Working and answer

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# Practice workbook · Write a logic statement for a problem

Turning a written set of conditions into a Boolean expression using AND, OR, NOT and brackets.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-11

# 1. Question 5 · part d.ii · 3 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part d.ii only.

**5** Hardware devices execute programs to carry out a variety of tasks.

(a) A program controls a bee character in an animation.

The bee can turn to face North, East, South or West.

The bee can move any number of steps in the direction it is facing.

Complete the table to show one input and one output required to move the bee.

**(2)**

| Requirement | Example |
|---|---|
| Input | |
| Process | Calculate the path the bee will move along to its new position |
| Output | |

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# 1. Question 5 · part d.ii · 3 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part d.ii only.

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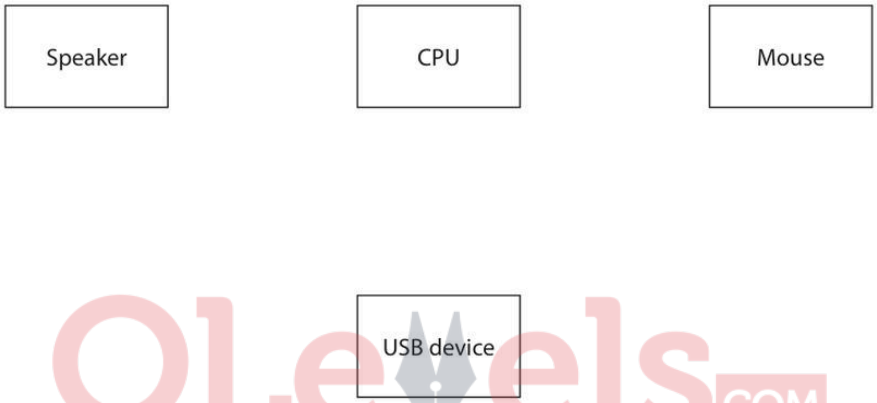
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(b) Computers are made up of hardware components.

(i) Complete the diagram by adding directional arrows between the components to show the flow of communication.

**(3)**




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# 1. Question 5 · part d.ii · 3 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part d.ii only.

(ii) Identify what is stored in ROM.

(1)

- ☒ **A** The software firewall
- ☒ **B** The basic input output system
- ☒ **C** The operating system
- ☒ **D** The user interface code

(iii) Cache is used as temporary storage.

One type of cache is located between main memory and the CPU.

Explain **one** reason cache is used in a computer.

(2)

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# 1. Question 5 · part d.ii · 3 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part d.ii only.

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(c) A washing machine uses several different embedded systems.

One embedded system uses a switch to identify the type of wash cycle selected by the user.

Describe **one other** example of an embedded system found in a washing machine.

**(2)**

(d) Programs use logic statements to control physical hardware.

(i) A window shuts when the temperature is too cool or it is a rainy night.

The values are defined as:

- A shows it is night time
- B shows it is too cool
- C shows it is raining.

Complete the truth table to show the results of each operation.

Two rows have been done for you.

**(3)**

| A | B | C | A AND C | (A AND C) OR B |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | | |
| 0 | 1 | 0 | | |
| 0 | 1 | 1 | | |
| 1 | 0 | 0 | | |
| 1 | 0 | 1 | | |
| 1 | 1 | 0 | | |
| 1 | 1 | 1 | 1 | 1 |

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P 7 2 9 3 7 A 0 1 7 2 4

17

*Turn over* ▶

# 1. Question 5 · part d.ii · 3 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part d.ii only.

(ii) A warehouse has an automated alarm system.

When the alarm system is activated it will sound if:

- a movement sensor (M) is activated
- a pressure pad (P) is activated
- a key code (C) to deactivate the alarm system has not been entered.

Construct a logic statement, using AND, OR and NOT with the letters M, P and C, to show the conditions that will sound the alarm.

**(3)**

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# 1. Question 5 · part d.ii · 3 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part d.ii only.

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(e) A computer with a single CPU runs several processes at the same time.

This computer is multitasking.

Describe how the operating system enables processes to share a single CPU.

(2)

**(Total for Question 5 = 18 marks)**

Working and answer

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## 2. Question 5 · part b · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

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**5** Programmers work as part of a team to develop software.

(a) They use truth tables to determine if an expression evaluates to true or false.

Complete this truth table for the Boolean expression

**P AND (Q OR R)**

You may not need to use all rows.

**(3)**

| P | Q | R | Q OR R | P AND (Q OR R) |
|---|---|---|---|---|
| | | | | |
| | | | | |
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## 2. Question 5 · part b · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

(b) The team is working on software for a ride-on lawn mower.

The lawn mower must not start unless it is safe for the driver.

- The brake (B) must be engaged.
- The driver's seat sensor (S) must be engaged.
- The blade safety lock (L) must be removed.

Construct a Boolean expression, using AND, OR, and NOT with the letters B, S and L to show the safe starting condition.

**(2)**

(c) The team uses code reviews as part of the software development process.

State **two** ways in which code reviews can help protect software systems from cyber attacks.

**(2)**

1 .....

2 .....

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## 2. Question 5 · part b · 2 marks · Working and answer

Working and answer

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## 3. Question 6 · part c · 4 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.

**6** Akiko develops software programs.

(a) Akiko can use a low-level or a high-level programming language to write code.

(i) Give **one** benefit of using a low-level programming language to develop a program.

**(1)**

.....

.....

(ii) Give **one** drawback of using a low-level programming language to develop a program.

**(1)**

.....

.....

(b) Akiko is writing an anti-virus program.

(i) The program will include a virus removal function.

Give **two other** functions that should be included.

**(2)**

1 .....

.....

2 .....

.....

(ii) An anti-virus program is one type of anti-malware.

Give **one other** type of anti-malware.

**(1)**

.....

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## 3. Question 6 · part c · 4 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.

(c) Akiko is also working on a burglar alarm system.

The alarm must only be triggered if:

- the alarm (A) is set
- it is dark outside (O)
- a window (W) or a door (D) has been opened.

Construct a Boolean expression, using AND, OR and NOT with the letters A, O, W, and D to show the conditions that will trigger the alarm.

**(4)**

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## 3. Question 6 · part c · 4 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.

(d) Akiko wants to know how the software she writes can be protected.

Discuss how the software can be protected from illegal copying and/or illegal distribution.

You should consider:

- intellectual property
- licensing.

**(6)**

**(Total for Question 6 = 15 marks)**

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## 3. Question 6 · part c · 4 marks · Working and answer

Working and answer

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# 4. Question 5 · part e · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part e only.

**5** A team is developing a computer game.

(a) Complete the table by adding a tick (✓) to show the characteristics of program language translators.

You may select more than one translator per characteristic.

**(3)**

| Characteristic | Translator | | |
|---|---|---|---|
| | **Assembler** | **Interpreter** | **Compiler** |
| Does **not** have to be present in memory to execute the translated code. | | | |
| Translates high level language source code. | | | |
| Executes high level code until either the program has finished, or a syntax error is encountered. | | | |
| Translates low level language source code into machine code. | | | |

(b) Luca is testing the computer game.

The computer that he is using is an older model with a processor that runs at a low clock speed.

Explain **one** drawback of testing the game using this computer.

**(2)**

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P 7 5 7 3 6 A 0 1 8 2 8

## 4. Question 5 · part e · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part e only.

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(c) Anna uses utility software.

Complete the table by adding the type of utility software she should use to solve each issue.

| Issue | Utility software |
|---|---|
| Her word processor cannot read all of a file because it is corrupt. | |
| She has a JPEG image and needs to send it to a publisher who can only read PNG image files. | |
| There has been a flood and files on a server have been damaged, but Anna needs to access her data. | |
| Her hard drive is running very slowly when she tries to access a file. | |

(4)

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# 4. Question 5 · part e · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part e only.

(d) Tarik is writing an instruction manual for the computer game.

He decides to use software to compress the instruction manual before emailing it to a proofreader for checking.

State and justify the most appropriate compression method.

(3)

Method

.....

Justification

.....

.....

.....

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(e) Members of the team developing the game are either coders or designers.

Coders (C) write the application code.

Designers (D) design the graphics for the game and write instruction manuals.

Team members have one role only.

Construct a logic statement, using AND, OR and NOT with the letters C and D, to show the conditions for being a member of the team.

(3)

**(Total for Question 5 = 15 marks)**

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## 4. Question 5 · part e · 3 marks · Working and answer

Working and answer

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